AIX-MARSEILLE UNIVERSITY

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FACULTY OF ECONOMICS AND MANAGEMENT

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AIX MARSEILLE SCHOOL OF ECONOMICS (AMSE)

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MASTER'S DEGREE IN ECONOMETRY AND STATISTICS, DATA SCIENCE TRACK

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FINAL PROJECT

Predictive modeling for energy infrastructure optimization : analysis of charging station status and forecasting of household electricity consumption

Presented and written by :

Isaac BESSANH

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Introduction

The energy transition is profoundly transforming the way electrical infrastructures are used, managed, and optimized. In this context, the rise of electric vehicles and the evolution of household consumption patterns make it essential to develop smarter and more anticipatory grid management. This project fits within this dynamic by exploring two complementary challenges: predicting the operational status of electric vehicle charging stations and forecasting household electricity consumption.

These two dimensions fall under the same overarching objective: optimizing the management of energy networks in a context of transition toward systems that are more sustainable, smarter, and increasingly in demand. The choice of this topic is justified by the growing importance of charging infrastructures in electric mobility and by the need to anticipate household energy demand to ensure grid stability. By combining these two datasets, the project adopts an integrated approach to energy optimization.

The reliability of charging stations conditions the adoption of electric vehicles: predicting their status helps improve maintenance, availability, and user trust. At the same time, forecasting household electricity consumption is essential to avoid overloads, optimize production, and integrate more renewable energy. By combining these two issues, the project contributes to a comprehensive vision of energy optimization, where electric mobility and residential consumption are treated as interdependent components of a single system.

Chapter 1

Prediction of the status of electric vehicle charging stations

1.1 Data description

In order to predict the status of electric vehicle charging stations in different countries, we opted for a cross-section database containing data on electric vehicle (EV) charging stations worldwide, extracted from the OpenChargeMap public API in November 2025, which we downloaded from kaggle. It focuses on key details of each station, including location, operator, status, and connector types. The data is ideal for infrastructure planning and predictive modeling.

Variables	Descriptions	Types
id	Unique station ID	Integer
title	Station name (e.g., "Electra - Wambrechies")	String
address	Street address	String
town	City or town	String
state	State or province (may be blank)	String
postcode	ZIP / postal code	String
country	ISO 3166-1 alpha-2 country code (e.g., FR)	Categorical
lat	Latitude (WGS84)	Continuous numeric
lon	Longitude (WGS84)	Continuous numeric
operator	Charging network (e.g., Tesla, Electra)	Categorical
status	Operational status (e.g., "Operational", "Not Operational")	Categorical
num_connectors	Number of charging plugs	Integer
connector_types	Plug types (e.g., "CCS (Type 2)|Type 2")	Categorical
date_added	Date when station was added (UTC)	Datetime

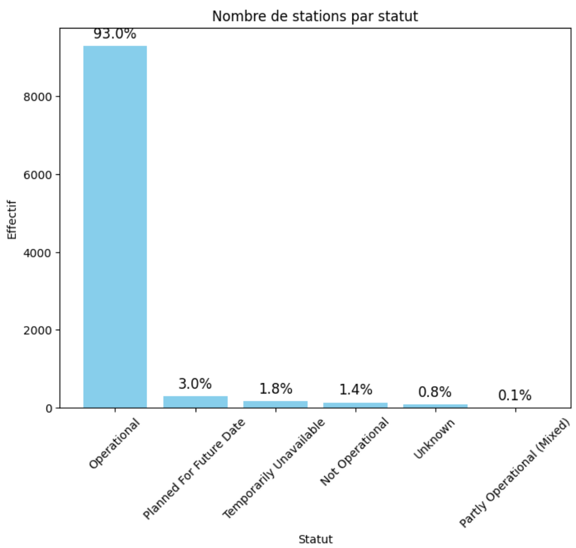
Table 1.1 – Description of the variables in the charging station dataset

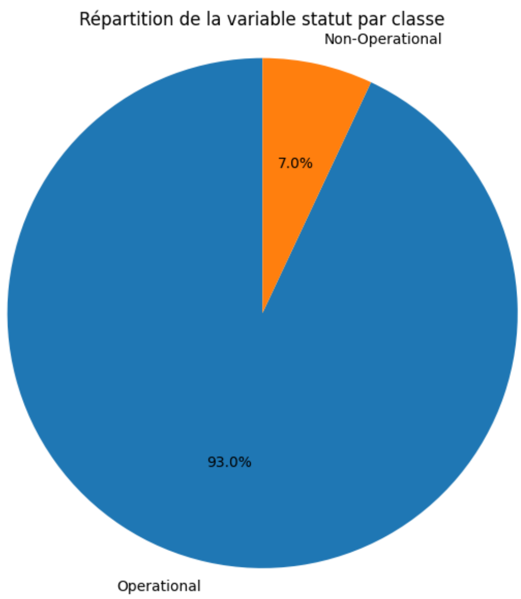
1.2 Exploratory analysis and data preparation

Our database contains 10,000 observations and 14 columns. It contains no duplicates, and missing values have been replaced with "unknown" to preserve all information. An exploratory analysis was performed to observe the distribution of each variable across different categories. The categories of some variables were grouped to create a clean basis for modeling, as is the case for the following variables, for example:

1.2.1 Status analysis

This variable, which had six categories, was grouped into two categories.

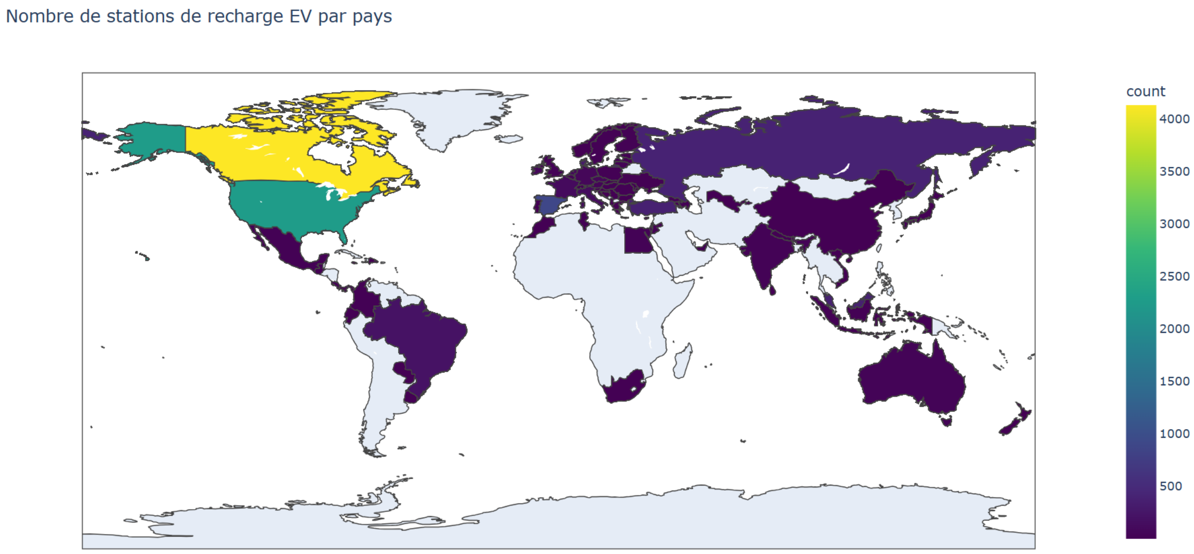




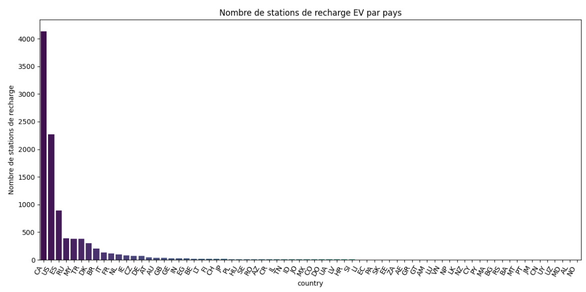
We can therefore conclude that only 7% of the stations are non-functional, compared to 93% that are functional. This represents an imbalance between our two categories, which will be taken into account during the modeling process.

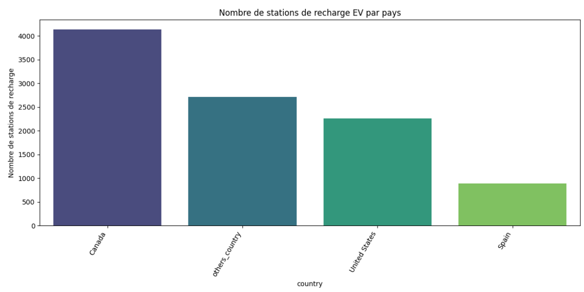
1.2.2 Country analysis

Regarding the country, we have 72 different countries in our database with countries that have few or no charging stations.



We therefore proceeded to group the countries according to their distribution. Countries with a high number of stations were retained, and the remaining countries with a very low number or none at all were grouped into one category.

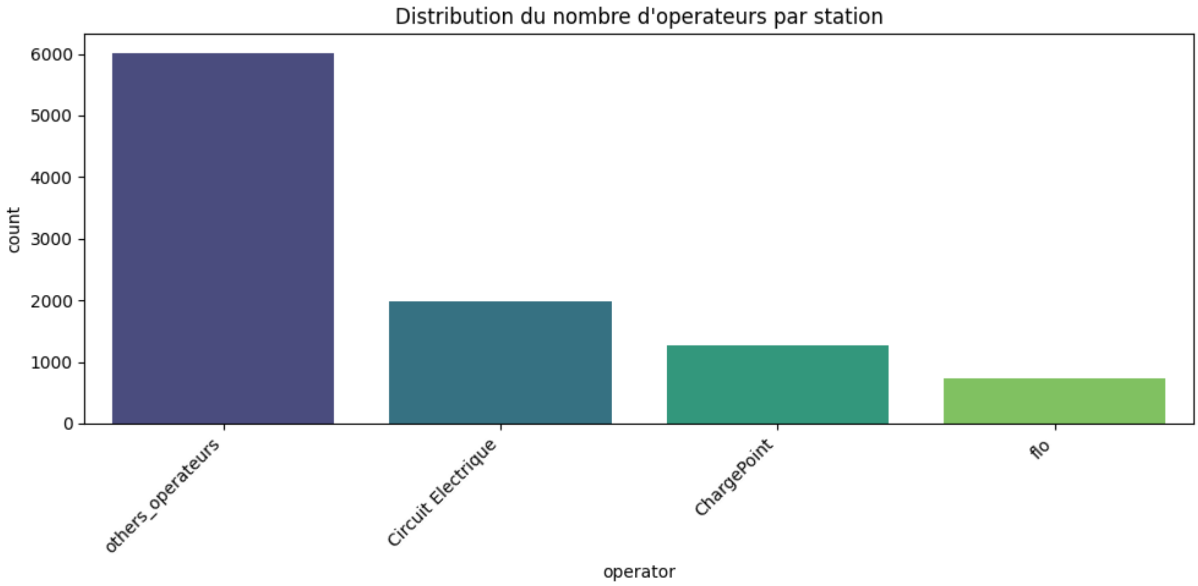




The graph highlights a strong heterogeneity in the distribution of charging stations for electric vehicles, with Canada dominating, followed by the United States, while Spain shows a significantly lower number of stations.

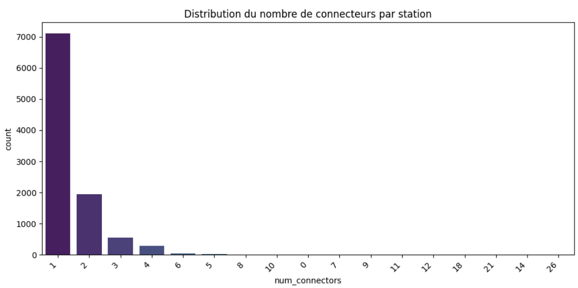
1.2.3 Operator analysis

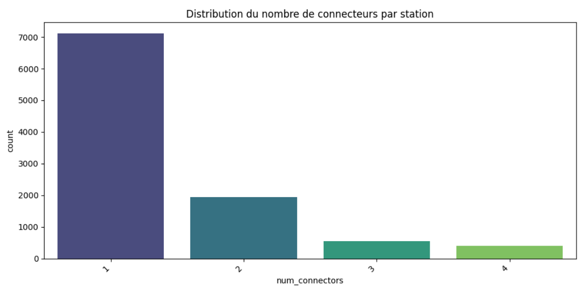
We also have a very high number of operators, namely 264, which we have grouped into four categories of operators.



1.2.4 Num-connectors analysis

The number of connectors has been reduced to 4 with the fourth category which includes all stations from four to twenty-six connectors.

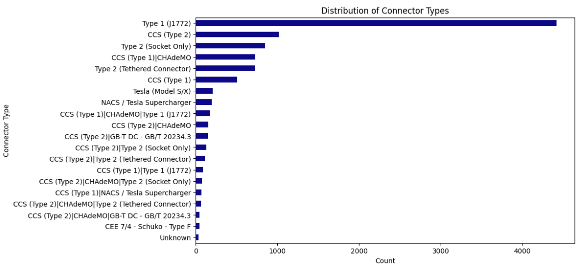


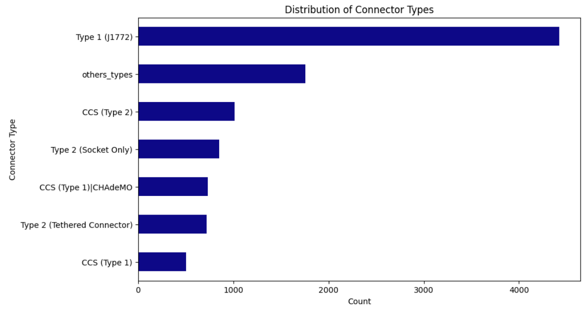


The distribution of the number of connectors per station reveals a high concentration of low-capacity infrastructure, indicating a deployment strategy based on the multiplication of small stations rather than the development of high-capacity charging hubs.

1.2.5 Connector-types analysis

Our database also contains a large number of connector types, 71 in total, which we have ultimately grouped into seven categories.





The distribution of connector types reveals a strong dominance of the Type 1 standard, reflecting technological inertia and an incomplete transition to fast charging standards, which are nevertheless essential to the growth of electric mobility.

1.3 Selection and construction of the target variable

Our database contains data on electric vehicle charging stations in different countries. We chose the status of the charging stations as our target variable, transforming it into a binary variable indicating whether a station is operational or not.

- Class 1 for stations considered available and functional,

- Class 0 for stations that are out of service, out of order, or unavailable.

This construction allows us to formulate the problem as a binary supervised classification, suitable for machine learning methods.

1.3.1 Importance of choice

This variable is crucial because it directly reflects the perceived reliability of the charging network from the perspectives of both users and operators. For an electric vehicle driver, knowing whether a station is likely to be operational influences trip planning and reduces range anxiety. For infrastructure managers, predicting the status allows them to anticipate outages, prioritize maintenance, and improve service quality. From a broader perspective, the actual availability of charging stations is a key driver of the energy transition, as an unreliable network hinders the adoption of electric vehicles.

1.3.2 Definition and justification

We define the variable ‘status‘ as a binary indicator of the operational availability of a charging station at a given time. This definition is based on a reasoned grouping of the categories in the dataset. This choice is justified both practically and methodologically:

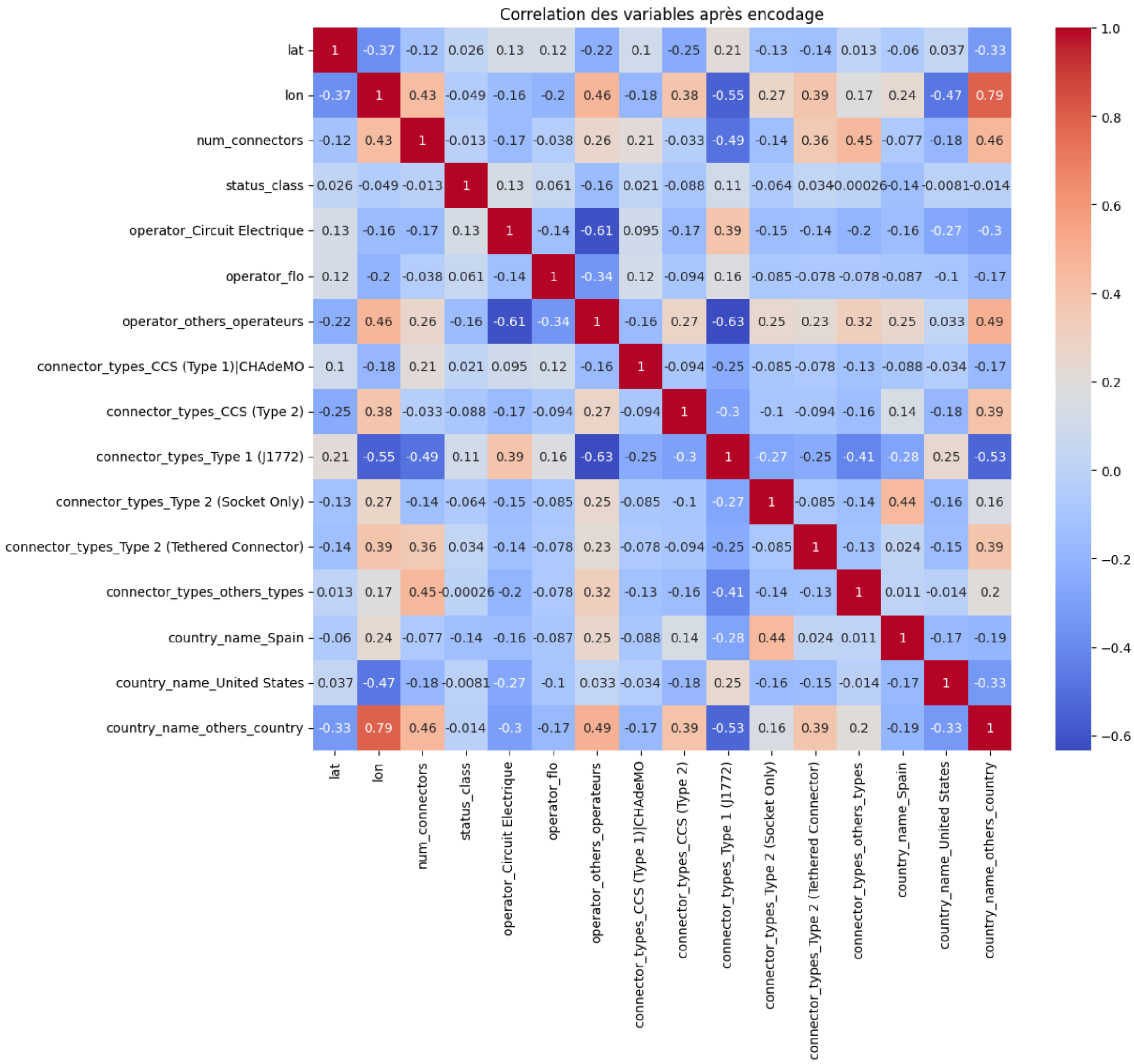
- practically, it corresponds to the real question a user asks: “Can I rely on this charging station or not?”;

- methodologically, it simplifies the problem into a binary classification task, which allows for clear comparison of different models, management of class imbalance, and interpretation of the results in terms of the probability of proper functioning.

Thus, the variable ‘status‘ condenses complex information into an indicator directly usable for operational decision-making.

1.4 Choice of explanatory variables

To predict the operational status of charging stations, we selected a set of explanatory variables from the initial dataset, such as "operator," "connector-types," "country-name," "num-connectors," "lat," and "lon". They were then encoded and transformed to obtain only numerical variables compatible with machine learning models (one-hot encoding for categorical variables, conservation or normalization for continuous variables). We thus obtained 10,000 observations and 15 explanatory variables after encoding.



The correlation matrix shows the correlations between encoded variables related to electric vehicle charging stations, revealing geographical, technical, and operator-country relationships. We observe that certain connector types are strongly associated with specific countries (e.g., Type 2 with Spain, Type 1 with the

United States), that the number of connectors is positively correlated with certain technical standards. These correlations allow us to identify redundancies, guide the selection of relevant variables for modeling.

1.5 Rebalancing of the target variable using SMOTE

After standardizing the variables, applying SMOTE to the training data corrected a significant imbalance between the classes. This balancing improves the model's ability to learn the characteristics of the rare class and ensures more reliable performance, particularly on metrics sensitive to imbalance such as recall and the F1 score.

Classe	**Before SMOTE**	**After SMOTE**	**Probability**
1	6976	6976	50%
0	524	6976	50%

Table 1.2 – Distribution of classes before and after SMOTE

1.6 Modeling

To better optimize our models, we applied a variable selection technique (Elastic Net) to the linear and logistic regression models. For models such as Random Forest and XGBoost, we applied cross-validation.

After encoding, variables were removed to avoid multicollinearity.

Variable	**Category removed (reference)**
operator	ChargePoint
connector_types	CCS (Type 1)
country_name	Canada

Table 1.3 – Categories removed during one-hot encoding

1.6.1 Linear Regression with Elastic Net

The combined application of SMOTE and Elastic Net penalized linear regression shows that the model judged only one variable, "lon," as irrelevant for prediction, as it was the only one eliminated from the initial 15 variables. The fact that 14 variables were retained indicates that the vast majority of predictors make a useful contribution to the model, even after class rebalancing by SMOTE.

So we have:

Data	**Value**
Train_data (after SMOTE)	(13952, 14)
Test_data	(2500, 14)

$$Y_i = X_i'\beta + \varepsilon_i$$

$$\begin{aligned}\text{status\_class}_i = {} & \beta_0 + \beta_1\,\text{lat}_i + \beta_2\,\text{num\_connectors}_i + \beta_3\,\text{operator\_CircuitElectrique}_i \\ & + \beta_4\,\text{operator\_flo}_i + \beta_5\,\text{operator\_others\_operateurs}_i + \beta_6\,\text{connector\_CCS1\_CHAdeMO}_i \\ & + \beta_7\,\text{connector\_CCS2}_i + \beta_8\,\text{connector\_Type1\_J1772}_i + \beta_9\,\text{connector\_Type2\_Socket}_i \\ & + \beta_{10}\,\text{connector\_Type2\_Tethered}_i + \beta_{11}\,\text{connector\_others\_types}_i \\ & + \beta_{12}\,\text{country\_Spain}_i + \beta_{13}\,\text{country\_UnitedStates}_i + \beta_{14}\,\text{country\_others\_country}_i + \varepsilon_i\end{aligned} \tag{1.1}$$

- Theoretical restrictions on errors

The following three assumptions must be met:

1- Conditional expectation zero

$$\mathbb{E}[\varepsilon_i \mid X_i] = 0$$

2- Constant conditional variance (homoscedasticity)

$$\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$$

3- Independence (or at least non-correlation) of errors

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0 \text{ pour } i \neq j$$

For OLS estimation, it is not necessary to assume a parametric distribution for ε_i in order to obtain unbiased and consistent estimators. For inference (tests and confidence intervals), we also adopt the usual assumption of conditional normality $\varepsilon_i \mid X_i \sim \mathcal{N}(0, \sigma^2)$.

- Estimate

The model explains approximately 23% ($R^2 = 0.232$) of the variance in terminal status, which remains modest but consistent with a complex phenomenon dependent on numerous unobserved operational factors.

Variable	**Coef.**	**Std. Err.**	**t**	**P>|t|**	**IC 95%**	
					0.025	**0.975**
const	0.6001	0.004	149.298	0.000	0.592	0.608
lat	0.0084	0.004	2.087	0.037	0.001	0.016
num_connectors	-0.0597	0.006	-9.691	0.000	-0.072	-0.048
operator_CircuitElectrique	0.1727	0.007	24.905	0.000	0.159	0.186
operator_flo	0.0870	0.006	15.219	0.000	0.076	0.098
operator_others_operateurs	-0.0929	0.007	-13.296	0.000	-0.107	-0.079
connector_types_CCS1_CHAdeMO	0.0471	0.006	8.451	0.000	0.036	0.058
connector_types_CCS2	0.0872	0.007	11.649	0.000	0.072	0.102
connector_types_Type1_J1772	0.0767	0.008	9.568	0.000	0.061	0.092
connector_types_Type2_Socket	0.0747	0.007	10.523	0.000	0.061	0.089
connector_types_Type2_Tethered	0.1731	0.007	23.762	0.000	0.159	0.187
connector_types_others_types	0.1604	0.008	20.116	0.000	0.145	0.176
country_Spain	-0.0756	0.006	-12.176	0.000	-0.088	-0.063
country_UnitedStates	0.0191	0.005	3.753	0.000	0.009	0.029
country_others_country	-0.0223	0.009	-2.397	0.017	-0.041	-0.004

Table 1.4 – Results of the linear regression with Elastic Net and 95% confidence intervals

The significant coefficients indicate that certain operators and connector types increase the probability of being operational, while variables such as the number of connectors or certain alternative operators are associated with a lower probability, revealing structural dynamics in infrastructure reliability. We have for exemple:

- `lat (coef = 0.0084)`: Latitude has a small but significant positive effect, suggesting that charging stations located further north are slightly more likely to be operational.

- `num-connectors (−0.0597)`: The more connectors a charging station has, the less likely it is to be operational, which may reflect increased complexity or a higher load.

- `operator-Circuit Electrique` (+0.1727) and operator-flo (+0.0870): These operators are associated with a higher probability of proper operation, indicating better reliability or technical management.

These coefficients reflect a structure where reliability depends primarily on the operator, the connector type, and, to a lesser extent, the location. The effects are linear and additive, but their interpretation remains limited by the continuous nature of the model and the simplification it imposes on potentially nonlinear relationships.

- Prediction

The performance of the Elastic Net linear model shows a moderate ability to distinguish between operational and non-operational terminals, with a AUC of 0.74 indicating a correct but imperfect separation between the two classes.

Classe	Precision	Recall	F1-score	Support
0	0.12	0.75	0.21	175
1	0.97	0.58	0.73	2325
weighted avg	0.91	0.60	0.69	2500
Accuracy : 0.60				
AUC Score : 0.7415				

Table 1.5 – Classification model performance metrics

Regarding the overall performance of the model, we have:

Measure	Value
R^2 (test)	-2.2861
RMSE (test)	0.4625
Residual variance (σ^2)	0.1922
Standard deviation of the error	0.4384

Table 1.6 – Overall performance of the linear model

Index	Réel	Prob. Prédite	IC Lower	IC Upper
5030	1	0.326533	0.000000	1.0
8288	1	0.998153	0.138512	1.0
2657	1	0.354691	0.000000	1.0
4051	1	0.520865	0.000000	1.0
1303	1	0.442941	0.000000	1.0

Table 1.7 – Prediction intervals (95%) for 5 observations

The model strongly favors the majority class and remains limited in correctly predicting failed terminals.

The linear model exhibits moderate overall performance with a AUC of 0.7415, indicating a reasonable ability to distinguish operational from non-operational boundaries. However, its negative R^2 on the test (-2.2861) reveals poor generalization, suggesting that the model performs worse than a naive prediction based on the mean.

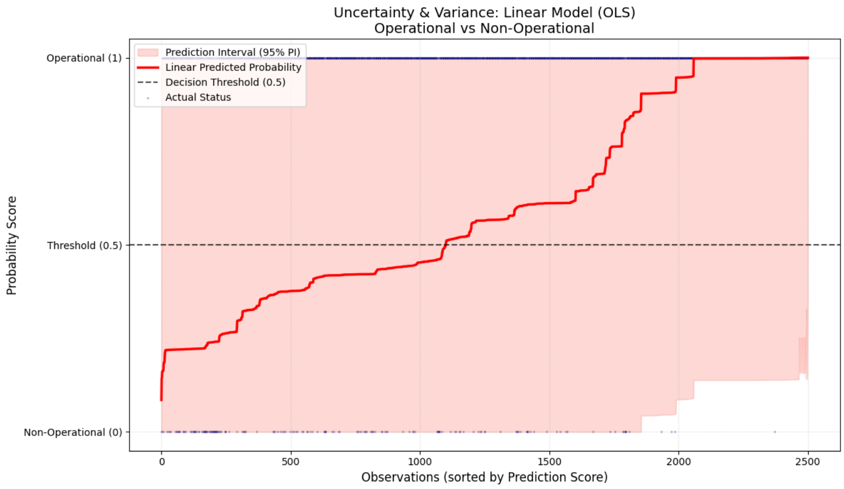


Figure 1.1 – OLS model results

The uncertainty plot of the linear model confirms the limitations already observed in the metrics: although the predicted probability curve follows an upward trend, the 95% prediction intervals are extremely wide and often cover the entire range from 0 to 1.

1.6.2 Logistical Regression with Elastic Net

The penalized logistic regression with Elastic Net retained 12 out of 15 variables, indicating that the model performed a relatively strict selection by eliminating only predictors with weak or redundant contributions.

We therefore obtain:

$$P(y_i = 1 \mid X_i) = \Lambda(X_i'\beta)$$

logistical variables kept	12 / 15
Eliminated variables (Elastic Net)	
connector_types_CCS_Type1_CHAdeMO lat lon	

Table 1.8 – logistical variables with Elastic Net

Unlike the linear model, the error is not directly modeled. However, an implicit error can be defined:

$$y_i = P(y_i = 1 \mid X_i) + \varepsilon_i$$

with

$$\varepsilon_i = y_i - P(y_i = 1 \mid X_i)$$

For observation i with feature vector $\mathbf{X}_i$, the probability of operational status is modeled as:

$$P(Y_i = 1 \mid \mathbf{X}_i) = \frac{1}{1 + e^{-(\omega_0 + \omega_1 X_{i1} + \cdots + \omega_k X_{ik})}}.$$

Here, $Y_i = 1$ indicates an operational station, and the parameters ω_j are estimated using optimization techniques.

so we have:

$$\begin{aligned}\log\left(\frac{P(status\_class_i = 1)}{1 - P(status\_class_i = 1)}\right) = \beta_0 + \beta_1\,\text{num\_connectors}_i + \beta_2\,\text{operator\_CircuitElectrique}_i \\ +\beta_3\,\text{operator\_flo}_i + \beta_4\,\text{operator\_others}_i + \beta_5\,\text{connector\_CCS2}_i \\ +\beta_6\,\text{connector\_Type1}_i + \beta_7\,\text{connector\_Type2Socket}_i \\ +\beta_8\,\text{connector\_Type2Tethered}_i + \beta_9\,\text{connector\_others}_i \\ +\beta_{10}\,\text{country\_Spain}_i + \beta_{11}\,\text{country\_US}_i \\ +\beta_{12}\,\text{country\_others}_i + \varepsilon_i\end{aligned} \tag{1.2}$$

- Theoretical restrictions on errors

In a logistic model, errors are not explicitly modeled as they are in a linear model. The theoretical assumptions are as follows:

1- Conditional expectation zero

$$\mathbb{E}[\varepsilon_i \mid X_i] = 0 \quad \text{(correct specification)}$$

2- Heteroscedastic conditional variance

$$\text{Var}(\varepsilon_i \mid X_i) = P_i(1 - P_i) \quad \text{(structural heteroscedasticity)}$$

3- Conditional independence of observations

$$\mathrm{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0 \quad \text{pour } i \neq j \quad \text{(independence of observations)}$$

4- Implicit parametric distribution (Bernoulli)

$$y_i \mid X_i \sim \mathrm{Bernoulli}(P_i) \quad \text{(replaces the normality assumption)}$$

-Estimate

Penalized logistic regression reveals a set of coefficients that clearly describe which factors increase or decrease the probability of a charging station being operational.

Variable	**Coef.**	**Std. Err.**	**z**	**P>|z|**	**IC 95%**	
					0.025	**0.975**
const	0.9834	0.057	17.248	0.000	0.872	1.095
num.connectors	-0.1929	0.028	-6.795	0.000	-0.249	-0.137
operator.Circuit_Electrique	1.8032	0.112	16.128	0.000	1.584	2.022
operator.flo	0.5324	0.039	13.629	0.000	0.456	0.609
operator.others_operateurs	-0.4078	0.033	-12.472	0.000	-0.472	-0.344
connector.types.CCS_Type2	0.3247	0.035	9.156	0.000	0.255	0.394
connector.types.Type1_J1772	0.2341	0.035	6.692	0.000	0.166	0.303
connector.types.Type2_Socket	0.2740	0.034	8.110	0.000	0.207	0.341
connector.types.Type2_Tethered	0.6792	0.034	19.814	0.000	0.612	0.746
connector.types.others_types	0.5946	0.036	16.500	0.000	0.524	0.663
country.name.Spain	-0.3558	0.031	-11.628	0.000	-0.416	-0.296
country.name.United_States	0.0701	0.024	2.863	0.004	0.022	0.118
country.name.others_country	-0.1384	0.045	-3.100	0.002	-0.226	-0.051

Table 1.9 – Results of the penalized logistic regression with 95% confidence intervals

The coefficients from the logistic regression after SMOTE and Elastic Net regularization allow us to identify the variables that influence the probability of a charging station's status being operational. We have for exemple:

- `num_connectors` (coef $= -0.1929$)

The number of connectors has a marked negative effect: the more connectors a charging station has, the less likely it is to be operational. This may reflect higher technical complexity or greater usage, increasing the risk of failure.

- `operator_Circuit_Electrique` (coef $= +1.8032$) This is one of the strongest effects of the model. Charging stations operated by Circuit_Electrique have a significantly higher probability of being operational, indicating particularly reliable technical management.

- `operator_flo` (coef = +0.5324) Charging stations operated by flo also have a higher probability of being functional, indicating good maintenance and operational quality.

- Prediction

The Elastic Net logistic regression performance shows a model that moderately separates the two classes, with an AUC of 0.7366 indicating decent but far from optimal discriminatory power. As with the linear model, the performance is highly asymmetrical: the majority class (operational bounds) is predicted with excellent accuracy (0.97) but average recall (0.58), while the minority class (non-operational bounds) suffers from very low accuracy (0.12) despite high recall (0.75).

Classe	Precision	Recall	F1-score	Support
0	0.12	0.75	0.21	175
1	0.97	0.58	0.73	2325
weighted avg	0.91	0.59	0.69	2500
Accuracy : 0.59				
AUC Score : 0.7366				
Mean variance of prediction : 0.000177				

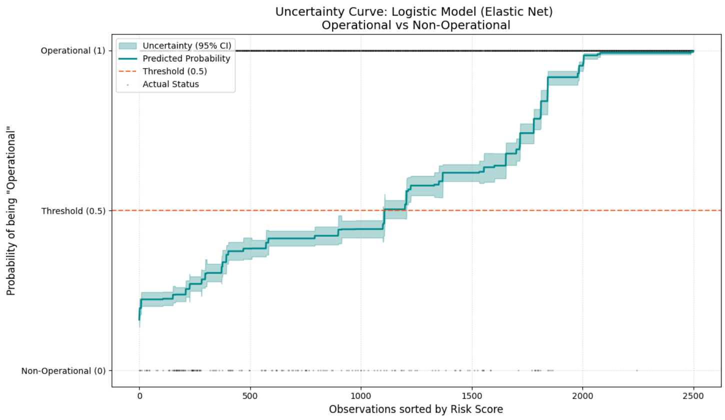
Table 1.10 – Classification model performance metrics

Regarding the overall performance of the model, we have:

Index	Réel	Prob. Prédite	IC Lower	IC Upper
5030	1	0.224110	0.203711	0.245921
8288	1	0.992146	0.986629	0.995397
2657	1	0.381359	0.355692	0.407706
4051	1	0.503362	0.474842	0.531861
1303	1	0.441754	0.414860	0.468995

Table 1.11 – Prediction intervals (95%)

With a mean variance of 0.000177, the Elastic Net logistic model demonstrates decent performance with an AUC of 0.7366, indicating a moderate ability to distinguish operational from non-operational boundaries. Its distinguishing feature is its low mean variance, reflecting high stability in its predictions. In summary, this model combines reasonable performance with excellent probabilistic reliability, making it more suitable for operational decision-making.



The uncertainty curve of the regularized logistic model (Elastic Net) shows a well-calibrated predictive structure, with increasing probabilities of success as observations are sorted by risk score. Unlike the linear model, the 95% confidence intervals are narrow and closely follow the probability curve, indicating well-controlled uncertainty.

1.6.3 Full logistic regression

For normal logistic regression without Elastic Net, the procedure is the same except that in this case we consider all variables. Since the variable "lon" is strongly correlated (0.79) with another variable, we removed it to avoid multicollinearity. We therefore obtain:

$$
\begin{aligned}
\log\left(\frac{P(status\_class_i = 1)}{1 - P(status\_class_i = 1)}\right) &= \beta_0 + \beta_1\,\mathrm{lat}_i + \beta_2\,\mathrm{num\_connectors}_i + \beta_3\,\mathrm{operator\_CircuitElectrique}_i \\
&+\beta_4\,\mathrm{operator\_flo}_i + \beta_5\,\mathrm{operator\_others\_operateurs}_i + \beta_6\,\mathrm{connector\_CCS1\_CHAdeMO}_i \\
&+\beta_7\,\mathrm{connector\_CCS2}_i + \beta_8\,\mathrm{connector\_Type1\_J1772}_i + \beta_9\,\mathrm{connector\_Type2\_Socket}_i \\
&+\beta_{10}\,\mathrm{connector\_Type2\_Tethered}_i + \beta_{11}\,\mathrm{connector\_others\_types}_i \\
&+\beta_{12}\,\mathrm{country\_Spain}_i + \beta_{13}\,\mathrm{country\_UnitedStates}_i + \beta_{14}\,\mathrm{country\_others\_country}_i + \varepsilon_i
\end{aligned} \tag{1.3}
$$

- Estimate

This full logistic regression performed after SMOTE and after removing `lon` due to its strong correlation (0.79) with another variable shows an overall consistent model where several technical, organizational, and geographical factors significantly influence the probability of a charging station being operational.

Variable	**Coef.**	**Std. Err.**	**z**	**P>|z|**	**IC 95%**	
					0.025	**0.975**
const	0.9784	0.057	17.271	0.000	0.867	1.089
lat	0.0362	0.019	1.891	0.059	-0.001	0.074
num_connectors	-0.2570	0.030	-8.444	0.000	-0.317	-0.197
operator_CircuitElectrique	1.7878	0.111	16.084	0.000	1.570	2.006
operator_flo	0.5085	0.039	13.117	0.000	0.433	0.584
operator_others_operateurs	-0.3774	0.033	-11.486	0.000	-0.442	-0.313
connector_types_CCS1_CHAdeMO	0.2021	0.030	6.702	0.000	0.143	0.261
connector_types_CCS2	0.4029	0.038	10.679	0.000	0.329	0.477
connector_types_Type1_J1772	0.3791	0.042	8.967	0.000	0.296	0.462
connector_types_Type2_Socket	0.3443	0.036	9.602	0.000	0.274	0.415
connector_types_Type2_Tethered	0.7689	0.037	20.735	0.000	0.696	0.842
connector_types_others_types	0.7219	0.041	17.758	0.000	0.642	0.802
country_Spain	-0.3452	0.031	-11.227	0.000	-0.405	-0.285
country_UnitedStates	0.0787	0.025	3.168	0.002	0.030	0.127
country_others_country	-0.1015	0.045	-2.254	0.024	-0.190	-0.013

Table 1.12 – Results of the full logistic regression with 95% confidence intervals

- Prediction

The performance of the full logistic model shows adequate discriminatory power but highly asymmetric behavior between the two classes. The AUC of 0.7413 indicates that the model distinguishes between operational and non-operational boundaries reasonably well, but the overall accuracy of 59% primarily reflects the difficulty in generalizing to a naturally unbalanced test set. In summary, even though the full model captures the data structure better than the penalized versions, it remains limited in effectively detecting faulty bounds, suggesting that nonlinear or more flexible approaches would be needed to improve fault detection.

Classe	Precision	Recall	F1-score	Support
0	0.12	0.75	0.21	175
1	0.97	0.58	0.73	2325
weighted avg	0.91	0.59	0.69	2500
Accuracy : 0.59				
AUC Score : 0.7413				
Mean prediction variance : 0.000205				

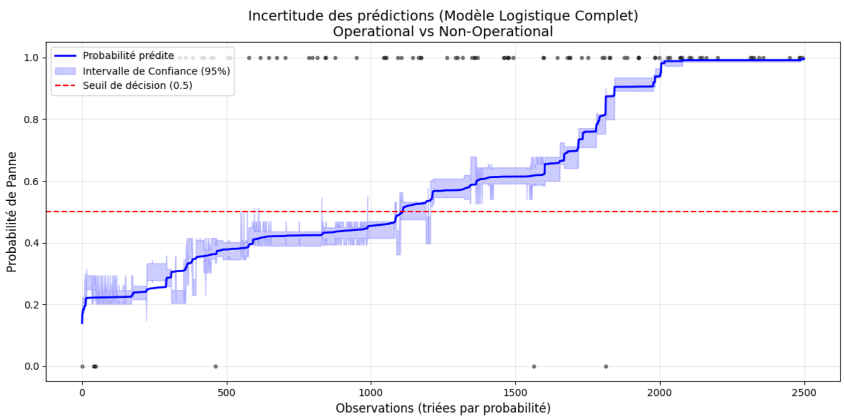
Table 1.13 – Classification model performance metrics

Regarding the confidence interval, we have:

Index	Réel	Prob. Prédite	IC Lower	IC Upper	Variance
5030	1	0.308121	0.273403	0.345153	0.000336
8288	1	0.991503	0.985591	0.995001	0.000005
2657	1	0.352959	0.322192	0.384995	0.000257
4051	1	0.525266	0.495298	0.555053	0.000233
1303	1	0.443678	0.415129	0.472605	0.000215

Table 1.14 – Prediction intervals (95%) for 5 observations

With a mean predicted variance of 0.000205, the full logistic model performs adequately with a AUC of 0.7413, reflecting a moderate ability to distinguish operational from non-operational bounds. Its low mean predicted variance (0.000205) is particularly noteworthy, indicating high stability in the estimates.



The uncertainty plot of the full logistic model shows a well-controlled predictive structure, with increasing probabilities of failure and narrow confidence intervals throughout the distribution. The operational and non-operational bounds are generally well separated, and the blue probability curve follows a smooth progression. This behavior reflects a good calibration of the model, capable of providing consistent and stable probabilistic predictions with low and well-quantified uncertainty.

1.6.4 Random Forest

We applied cross-validation to the training sample after SMOTE in order to optimize our model.

- Theoretical restrictions on errors

Random Forest is a nonparametric model that does not rely on any assumptions of normality, linearity, or homoscedasticity of errors. Observations must be independent, but the errors of individual trees can be correlated. The generalization error is theoretically bounded by the strength of the trees and their average correlation. In the presence of unbalanced classes, the errors are structurally biased toward the majority class. Finally, the error structure depends on the choices of depth, bootstrapping, and the number of variables selected at each node.

- Prediction

The Random Forest model trained after SMOTE offers a significant improvement over linear models, with an AUC of 0.7776 and, more importantly, a high accuracy of 85%, indicating that it captures the nonlinear structure of the data much better. The model performs particularly well in predicting operational (class 1) boundaries, with an accuracy of 0.96 and a recall of 0.88, reflecting an excellent ability to correctly identify functional stations.

Regarding the confidence interval, we have:

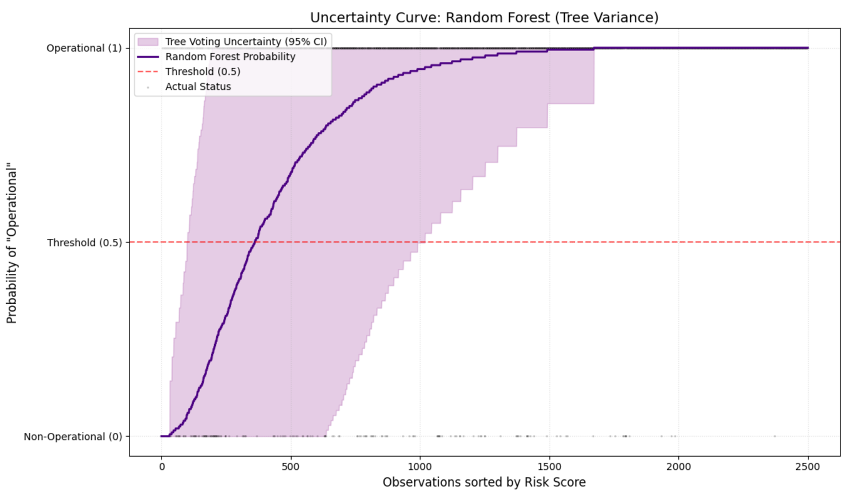
Classe	Precision	Recall	F1-score	Support
0	0.24	0.50	0.32	175
1	0.96	0.88	0.92	2325
weighted avg	0.91	0.85	0.88	2500
Accuracy : 0.59				
ROC AUC Score : 0.7776				
Mean variance of prediction : 0.062957				

Table 1.15 – Classification model performance metrics

Index	Réel	Prob. Prédite	IC Lower	IC Upper	Variance
5030	1	0.955	0.548684	1.000000	0.042975
8288	1	1.000	1.000000	1.000000	0.000000
2657	1	0.545	0.000000	1.000000	0.247975
4051	1	0.950	0.522828	1.000000	0.047500
1303	1	0.990	0.794982	1.000000	0.009900

Table 1.16 – Prediction intervals (95%) for 5 observations

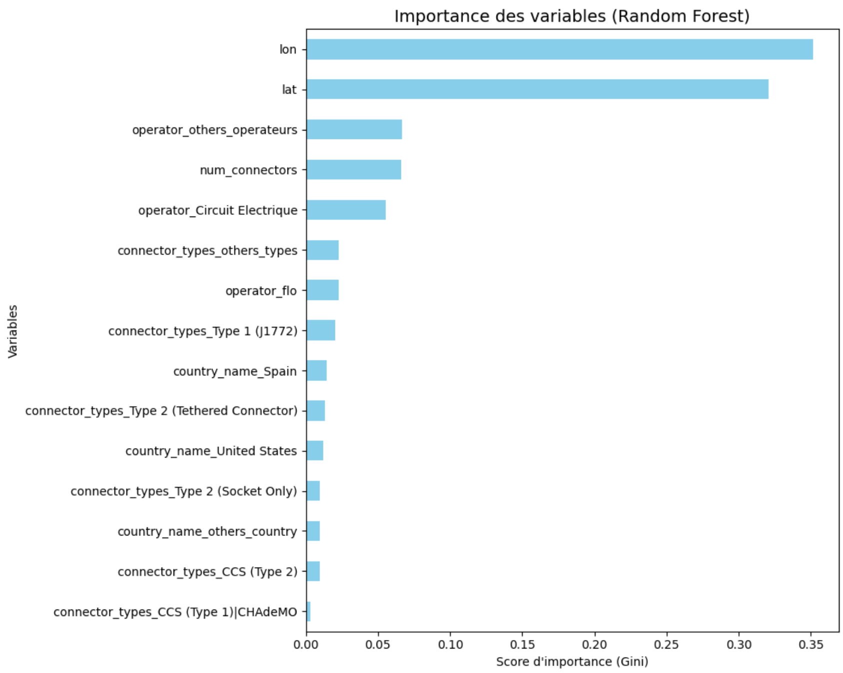
With a mean prediction variance of 0.062957, the Random Forest model demonstrates very good overall performance, with a ROC AUC of 0.7776 on the test and a mean AUC in cross-validation of 0.9598, indicating excellent discriminatory power and strong stability between folds. The Random Forest effectively captures the complex structures of the problem, but its probabilistic granularity is less reliable than that of logistic models, which may limit its interpretability for individual decisions.



The uncertainty curve of the Random Forest model reveals strong discriminatory power (AUC = 0.7776), but significant uncertainty in individual predictions. Although the average probabilities are high for the operational bounds, the 95% confidence intervals are often very wide, reflecting substantial variability between the model's trees.

- Importance of variables

Analyzing the relative importance of variables in the Random Forest model reveals the different variables that also influence the probability of being operational. This ranking highlights Random Forest's ability to capture the complex interactions between geography, operators, and technical configuration.



1.6.5 XGboost

We also applied cross-validation to the training sample after SMOTE in order to optimize our model.

- Theoretical restrictions on errors

Same for XGBoost, errors are not subject to the same theoretical constraints as in classical parametric models. These methods do not rely on any assumptions of normality, linearity, or homoscedasticity of the residuals, which makes them particularly flexible when dealing with complex data structures.

- Prediction

The XGBoost model, trained after class rebalancing using SMOTE, exhibits the best performance among the tested approaches, with a ROC AUC of 0.7946 and an overall accuracy of 80%, reflecting a good ability to discriminate between operational and non-operational boundaries. In summary, the SMOTE + XGBoost combination makes it possible to effectively capture complex interactions between variables and significantly improve the detection of non-operational boundaries, while maintaining high reliability on functional boundaries.

Classe	Precision	Recall	F1-score	Support
0	0.19	0.55	0.28	175
1	0.96	0.82	0.89	2325
weighted avg	0.91	0.80	0.84	2500
Accuracy : 0.80				
AUC Score : 0.7946				
Mean variance of prediction : 0.000530				

Table 1.17 – Classification model performance metrics

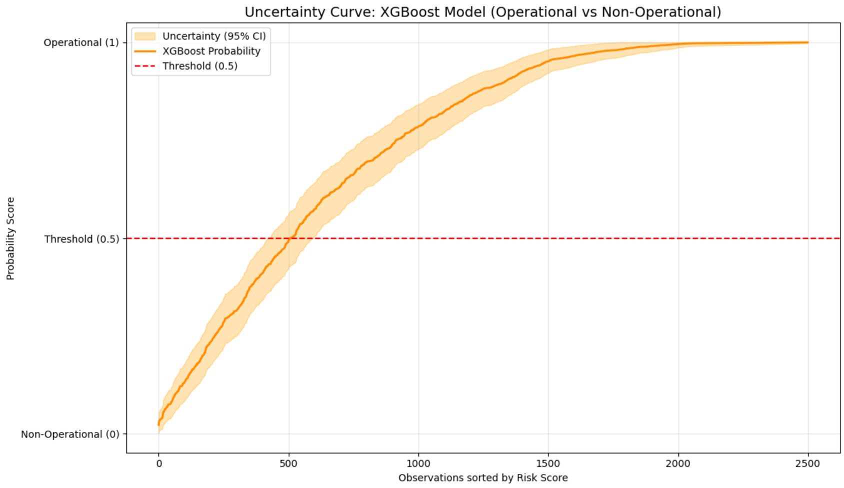
Regarding the confidence interval, we have:

Index	Réel	Prob. Prédite	IC Lower	IC Upper	Variance
5030	1	0.792973	0.736819	0.849127	0.000821
8288	1	0.992601	0.980723	1.000000	0.000037
2657	1	0.733786	0.672531	0.795041	0.000977
4051	1	0.554339	0.485453	0.623225	0.001235
1303	1	0.973086	0.950657	0.995515	0.000131

Table 1.18 – Prediction intervals (95%) for 5 observations

With a mean prediction variance of 0.000530, the XGBoost model stands out for its excellent performance and remarkable stability. With a AUC of 0.7946 on the test and a mean AUC of 0.9407 in cross-validation, it demonstrates a strong ability to discriminate between operational and non-operational bounds. XGBoost combines predictive power and probabilistic stability, making it a particularly suitable model for operational applications requiring both accuracy and confidence.

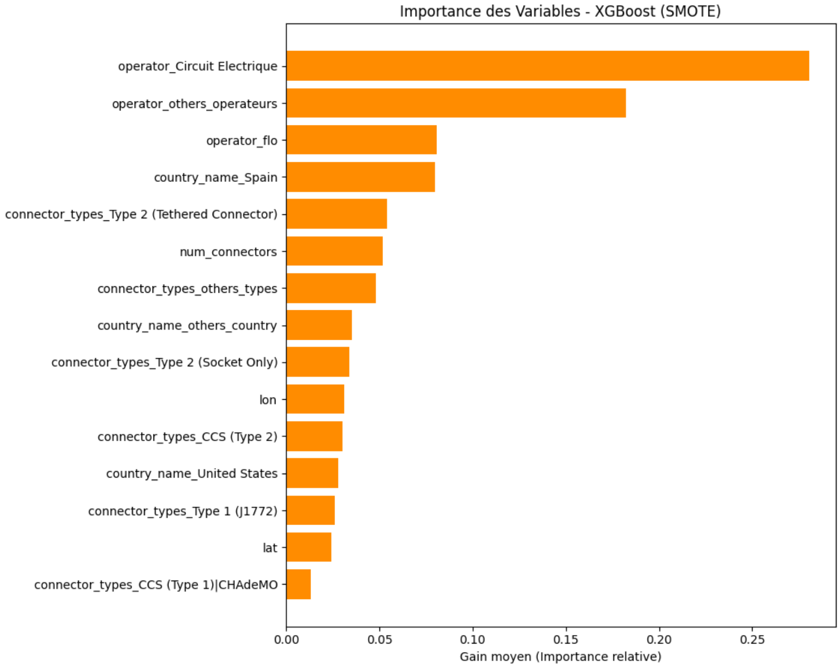
The uncertainty curve of the XGBoost model illustrates a



smooth and well-calibrated progression of operational probabilities, with a clear separation between operational and non-operational bounds. The 95% confidence intervals are very narrow, even for intermediate scores, demonstrating low and well-controlled uncertainty. In summary, XGBoost combines accuracy, robustness, and confidence, making it a particularly suitable tool for automated operational decisions.

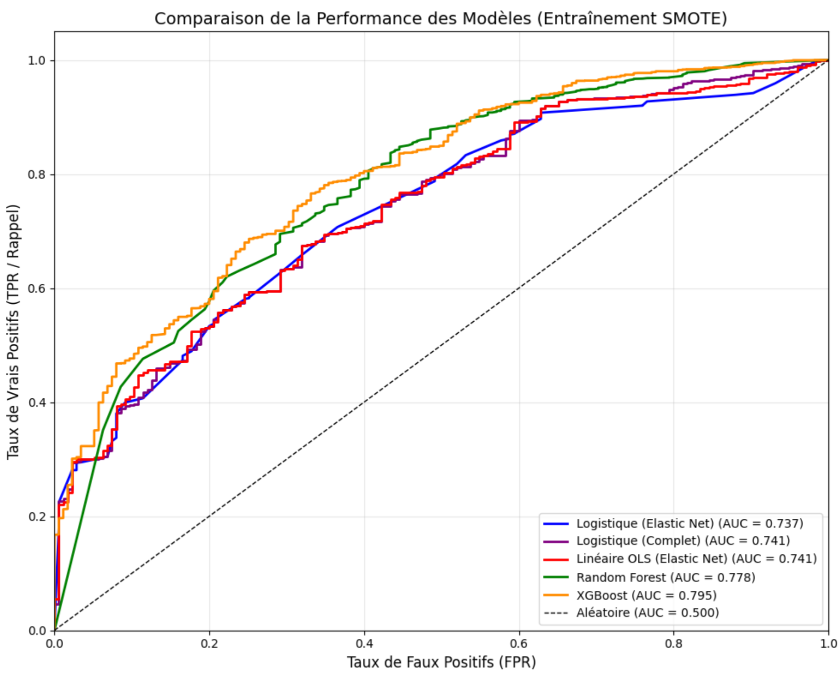
- Importance of variables

Analysis of variable importance in the XGBoost model trained with SMOTE reveals a predictive structure dominated by organizational and technical characteristics. This importance profile shows that XGBoost effectively leverages the interactions between operator, technology, and environment to model the status of terminals.



1.6.6 Choosing the best model

The XGBoost model trained with SMOTE stands out as the best performer overall, with an AUC of 0.7946 and an accuracy of 80%, demonstrating an excellent ability to discriminate between operational and non-operational terminals. In short, the SMOTE + XGBoost approach offers a robust, balanced model better suited to detecting faulty terminals in real-world contexts.



Chapter 2

Forecasting of household electricity consumption

2.1 Data Description

2.1.1 Context and Data Origin

In the context of forecasting domestic electricity consumption, we utilized a database from the UCI Machine Learning Repository titled *Individual Household Electric Power Consumption*. This database contains electricity consumption measurements from a household located in the Paris region, recorded at a frequency of one minute over a period of nearly four years (December 2006 to November 2010).

Analyzing domestic electricity consumption presents several major challenges in the current context of energy transition. On one hand, it allows grid managers to anticipate peak demand and optimize electricity production. On the other hand, it contributes to the integration of renewable energies by facilitating the balance between supply and demand. Finally, it offers consumers the opportunity to better understand and manage their energy consumption.

For this study, we focused on the period from **January 1st, 2007, to June 30th, 2007**—approximately six months of data—in order to model the short- and medium-term dynamics of electricity consumption. This choice of period allows us to capture both seasonal variations (the transition from winter to spring and then early summer) and weekly cycles related to the household's lifestyle habits.

2.1.2 Initial Data Structure

The initial dataset contains **260,640 observations** (minute-by-minute measurements) and includes the following variables:

Variable	**Description**	**Unit**
Date	Date of the observation	Date format
Time	Time of the observation	Time format
Global_active_power	Global active power	kilowatts (kW)
Global_reactive_power	Global reactive power	kilowatts (kW)
Voltage	Electric voltage	Volts (V)
Global_intensity	Global current intensity	Amperes (A)
Sub_metering_1	Sub-metering 1 (kitchen)	Watt-hour (Wh)
Sub_metering_2	Sub-metering 2 (laundry room)	Watt-hour (Wh)
Sub_metering_3	Sub-metering 3 (water heater/AC)	Watt-hour (Wh)

Table 2.1 – Description of the variables in the electricity consumption dataset

2.1.3 Preprocessing and Temporal Aggregation

The initial database contains measurements taken every minute, representing a very fine level of granularity. While this precision is useful for analyzing instantaneous behavior (turning on an appliance, consumption peaks), it also introduces a significant amount of noise and makes modeling more complex. Furthermore, for power grid management applications, daily consumption is often more relevant than minute-by-minute data.

We therefore proceeded with the following transformations to obtain usable data:

— **Merging temporal columns**: The `Date` and `Time` variables, initially separate, were combined into a single `Datetime` variable to precisely identify each observation in time.

- **Numerical conversion**: Consumption variables, initially stored as text, were transformed into numerical values (float64 type) to enable statistical calculations.
- **Handling missing values**: The dataset contained 3,771 observations with missing values, representing approximately 1.4% of the total. These rows were removed as they did not allow for reliable calculation. Given that this proportion is low, this removal does not significantly affect the quality of the analysis.
- **Daily aggregation**: The 260,640 minute-by-minute measurements were aggregated by day by calculating the **daily average** of the active power. This allows us to focus on weekly trends and cycles rather than sub-daily fluctuations. For example, instead of having 1,440 measurements for one day (24h $\times$ 60 minutes), we obtain a single value representing the average consumption for that day.

After these transformations, the final dataset contains **181 observations** (one per day over 6 months) and 7 numerical variables. This structure is well-suited for time-series modeling while retaining essential information regarding consumption patterns.

2.1.4 Target Variable and Justification

Among the seven variables available in the dataset, we selected the `Global_active_power` variable as our primary variable of interest for modeling. This variable represents the **average daily global active power**, measured in kilowatts (kW), and corresponds to the total electricity consumption of the household over a day.

This choice is justified by several practical and methodological reasons:

- **Global representativeness:** Unlike the sub-meters which only measure specific areas (kitchen, laundry, water heater), the global active power aggregates all electrical uses of the home. It therefore provides a complete view of energy demand.

— **Relevance for grid managers:** It is this total consumption that determines the load on the electrical grid. By forecasting this variable, we help operators anticipate demand and adjust production accordingly.

— **Rich temporal behaviors:** This variable captures seasonal variations (heating in winter, air conditioning in summer), days of the week (difference between working days and weekends), and times of day (morning and evening peaks).

— **Practical applicability:** Predicting this variable enables concrete applications: optimization of peak/off-peak pricing, management of self-consumption with solar panels, or the detection of anomalies in consumption habits.

The objective of the modeling is therefore to build a model capable of predicting the expected average electricity consumption for the coming days, based on the history of past consumption and taking into account observed regularities.

2.2 Exploratory Data Analysis

2.2.1 Raw Time Series Visualization

The first step in any time series analysis is to visualize the data to identify apparent structures. Figure 2.1 shows the daily evolution of global active power over the study period (January to June 2007).

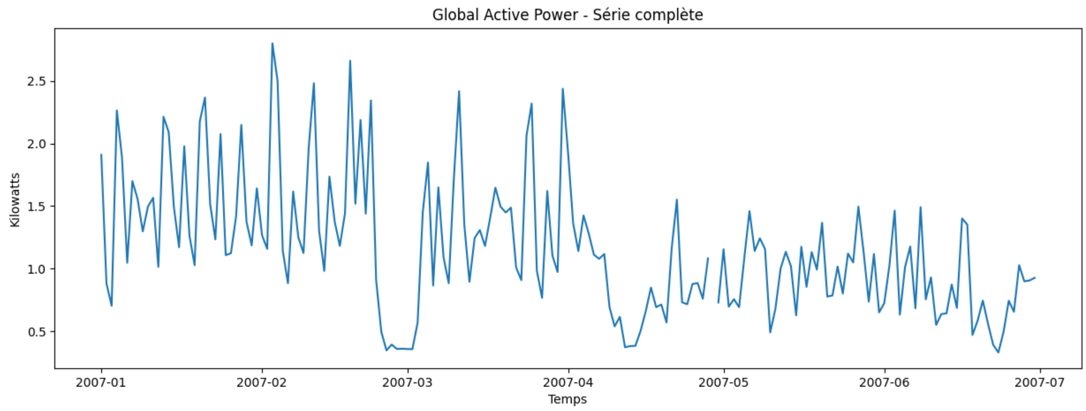


Figure 2.1 – Temporal evolution of global active power (kW) – Full series

This graphical representation allows us to make several important observations:

- **Significant volatility:** The series exhibits marked fluctuations from one day to the next, with values oscillating between 0.4 kW and 2.7 kW. These variations reflect the diversity of electrical usage depending on the day: some days require little energy (mild temperature, absence from home), while others see consumption climb (cold weather, prolonged presence, use of energy-intensive appliances).
- **Consumption peaks:** Daily peaks reaching up to 2.5 kW are observed, particularly at the beginning of the period (January-February). These peaks likely correspond to heating needs during the winter months, as well as the simultaneous use of several electrical appliances (oven, cooktop, washing machine, etc.).
- **Downward trend:** Looking at the overall picture, we notice that the average consumption level gradually decreases over the months. In January, average consumption is around 1.5 kW, while in June it drops toward 0.8–1.0 kW. This decline reflects the reduction in heating needs as spring and then summer temperatures set in.
- **Period of low consumption:** A sharp drop appears in March (values around 0.4 kW), which could be explained

by a prolonged absence of the occupants (vacation, business trip) or by particularly mild weather conditions reducing energy needs.

— **Regular oscillations:** Beyond the general trend, oscillations are visible that seem to repeat with a certain regularity. This structure suggests the presence of a weekly cycle, linked to differences in behavior between weekdays and weekends.

These initial observations indicate that the series is non-stationary (presence of a trend) and likely includes a seasonal component (weekly cycle). These elements will guide the choice of the forecasting model.

2.2.2 Seasonal Decomposition (STL)

To go beyond visual observation and precisely quantify the different components of the time series, we applied a statistical method called STL decomposition (*Seasonal and Trend decomposition using Loess*). This technique decomposes the observed series into three distinct elements: trend, seasonality, and residuals.

STL decomposition is based on the idea that any observation can be written as the sum (or product) of these three components:

$$\text{Observation}_t = \text{Trend}_t + \text{Seasonality}_t + \text{Residual}_t \tag{2.1}$$

For our analysis, we set the seasonal period to **7 days**, corresponding to the weekly cycle of electricity consumption habits.

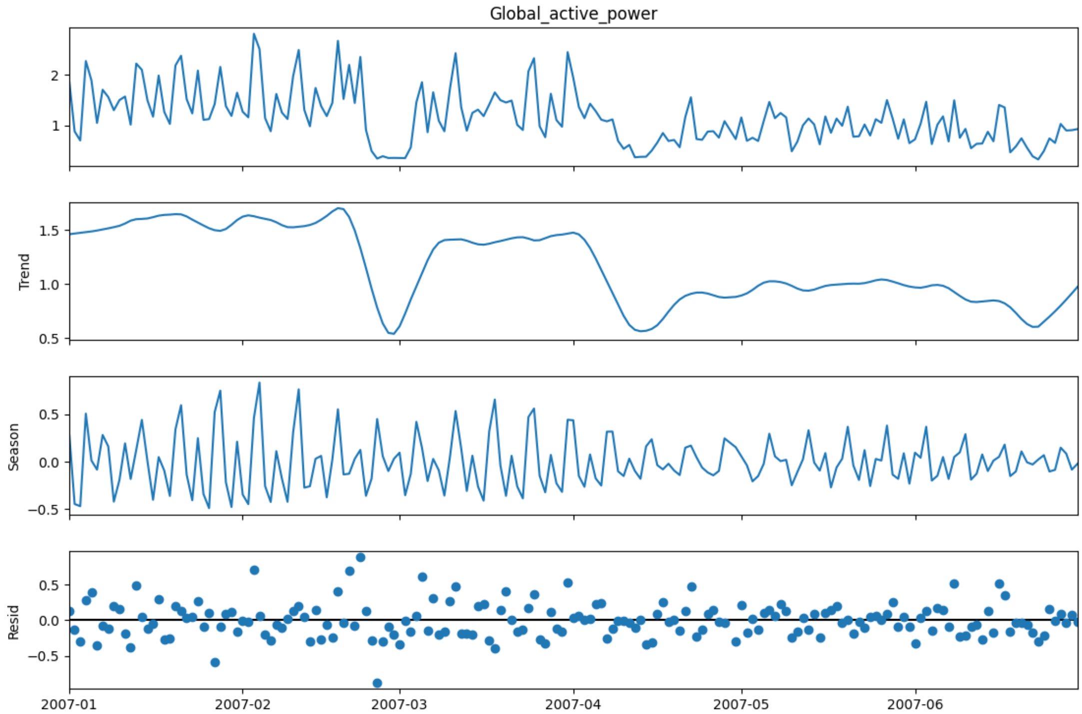


Figure 2.2 – STL decomposition of the `Global_active_power` series (period = 7 days)

Figure 2.2 presents four superimposed graphs, each revealing a different aspect of our data:

1. Observed Series : This is our raw data as recorded. It is the "full signal" containing all mixed phenomena: general trend, weekly variations, and random events.

2. Trend : This smooth curve represents the underlying evolution of electricity consumption, abstracting from short-term fluctuations. We clearly observe a progressive decrease from January (approx. 1.45 kW) to June (approx. 0.96 kW), representing a decrease of about 34% over six months.

This decrease is mainly explained by climatic factors: in winter, electric heating represents a significant portion of consumption. As temperatures rise in spring and early summer, this need decreases significantly. Other factors such as longer days (less need for artificial lighting) or changes in lifestyle habits (different cooking methods, less oven use) can also be imagined.

3. Seasonality : This graph reveals a pattern that repeats almost identically every week. The variations oscillate symmetrically around zero, with a relatively constant amplitude of approximately ± 0.5 kW.

This seasonal component reflects the **weekly habits** of the household. For example, consumption may be higher during the week (regular presence, use of household appliances) and lower on weekends (outings, different lifestyle). Or vice versa, depending on the household profile.

The fact that this cycle repeats very regularly indicates that the household's behaviors are predictable and structured over time. It is precisely this regularity that makes forecasting possible.

4. Residuals : This last component represents everything that could not be explained by the trend and seasonality. These are random, unpredictable, or exceptional events: an appliance failure, an unexpected cold snap, guests staying for several days, an unforeseen absence, etc.

The residuals fluctuate seemingly randomly around zero with no visible structure. This is a good sign: it means our decomposition successfully captured most of the structured information in the data.

2.2.3 Stationarity Analysis

A fundamental concept in time series analysis is **stationarity**. A series is said to be stationary when its statistical properties (mean, variance, autocorrelation) remain constant over time.

Why is stationarity important?

Most statistical models for time series, particularly the ARIMA family, rely on a fundamental assumption: **the rules governing the series remain the same over time**. In our case, we observed that the series has a downward trend; we must now **formally verify this** using a statistical test.

ADF Test on the original series

The Augmented Dickey-Fuller (ADF) test is the most common test for stationarity.

Test Result	**Value**
ADF Statistic	-1.973
p-value	0.298
Critical Values:	
1% Level	-3.470
5% Level	-2.879
10% Level	-2.576

Table 2.2 – ADF test results on the original series

Interpretation: The ADF statistic (-1.973) is **higher** than the three critical values. Furthermore, the p-value of 0.298 (29.8%) is well above the usual 5% threshold. **Conclusion:** We cannot reject the null hypothesis. The original series is **non-stationary**.

Stationarization by Detrending

A simple method to make a series stationary is to remove its trend component. We subtracted the STL trend from the original series:

$$\text{Detrended Series}_t = \text{Original Series}_t - \text{Trend}_t \tag{2.2}$$

Test Result	**Value**
ADF Statistic	-7.366
p-value	9.21×10^{-11}
Critical Values:	
1% Level	-3.469
5% Level	-2.879
10% Level	-2.576

Table 2.3 – ADF test results on the detrended series

Conclusion: The detrended series is **stationary**. We now have a series that meets the requirements for ARIMA or SARIMA modeling.

2.2.4 Autocorrelation Analysis (ACF and PACF)

We now analyze the temporal dependence structure using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF).

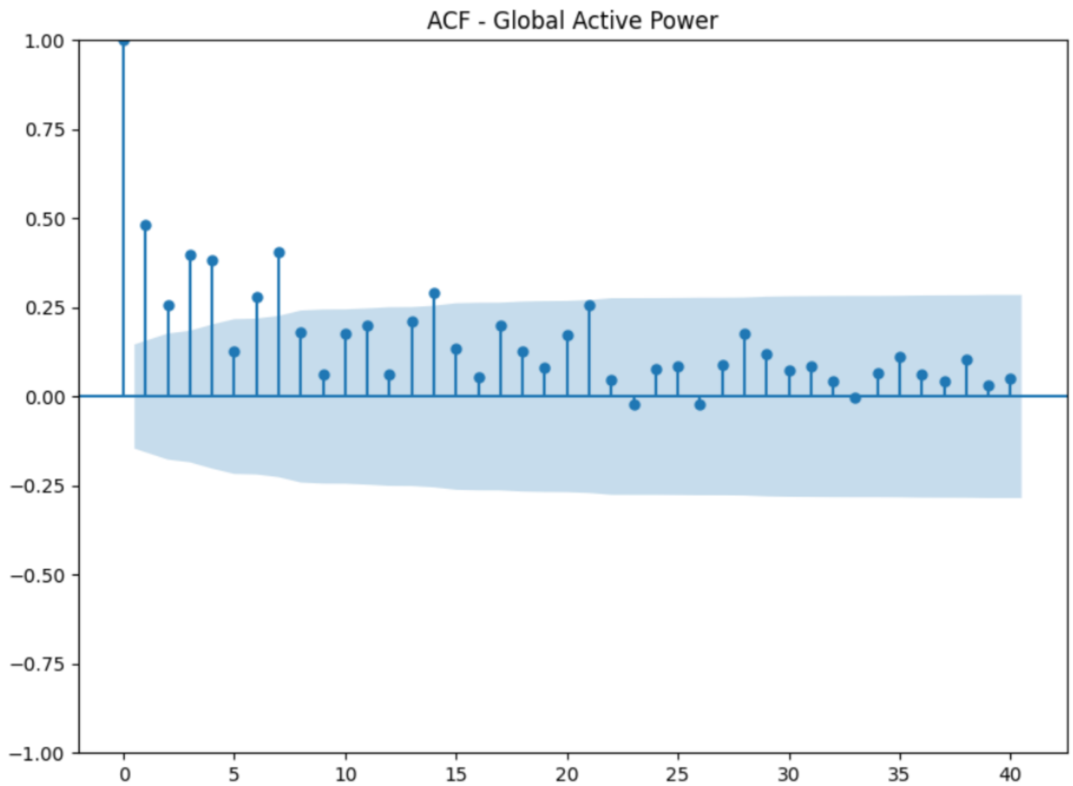


Figure 2.3 – Autocorrelation Function (ACF) of the `Global_active_power` series

Observations:

— **Spike at lag 1**: Indicates strong persistence; today's consumption is similar to yesterday's.

— **Gradual decay**: Characteristic of Moving Average (MA) processes.

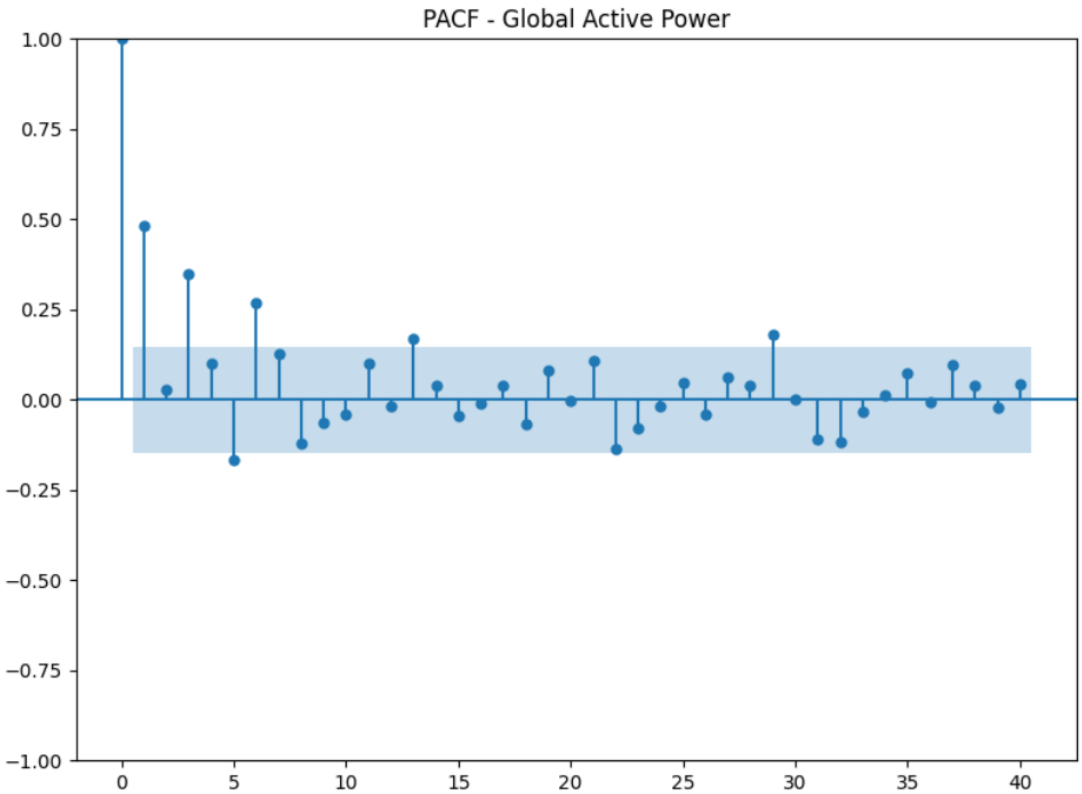


Figure 2.4 – Partial Autocorrelation Function (PACF) of the `Global_active_power` series

Observations:

— **Significant spike at lag 1**: Suggests a strong **autoregressive dependence**. Today's consumption directly depends on yesterday's.

Synthesis and Model Direction

Combining the ACF and PACF information:

— **ACF**: spike at lag 1 with decay → MA(1) component.

— **PACF**: spike at lag 1 → AR(1) component.

This points toward an **ARIMA(1,1,1)** model as a starting point. However, since we identified a weekly cycle, we will integrate a **seasonal component**, leading us to **SARIMA** models.

2.3 SARIMA Model Identification and Selection

2.3.1 What is a SARIMA Model?

The SARIMA (*Seasonal AutoRegressive Integrated Moving Average*) model is an extension of the classic ARIMA model that explicitly incorporates a seasonal component. It is one of the most powerful tools for forecasting time series that exhibit both short-term dynamics and recurring cycles.

Understanding the Components of an ARIMA Model

Before addressing SARIMA, let us briefly review the three components of an ARIMA(p,d,q) model:

1. AR (AutoRegressive) - Persistence : The AR component of order p models the idea that the current value depends on the p previous values. For example, in an AR(1) model, today's consumption is a function of yesterday's consumption:

$$y_t = \phi_1 y_{t-1} + \epsilon_t \tag{2.3}$$

where ϕ_1 is a coefficient to be estimated and ϵ_t is a random error term (white noise).

If ϕ_1 is close to 1, it signifies strong persistence: high consumption yesterday implies high consumption today. If ϕ_1 is close to 0, the past has little influence.

2. I (Integrated) - Trend Removal The parameter d indicates the number of times the series must be differenced to make it stationary. Differencing means calculating the changes from one period to the next:

$$\Delta y_t = y_t - y_{t-1} \tag{2.4}$$

A first-order differentiation (d=1) transforms a series with a trend into a series of variations. If the differenced series is still not stationary, a second differentiation is applied (d=2), and so on.

3. MA (Moving Average) - Memory of Shocks : The MA component of order q models the idea that the current value depends on the shocks (forecast errors) from the q previous periods. In an MA(1) model:

$$y_t = \epsilon_t + \theta_1 \epsilon_{t-1} \tag{2.5}$$

where θ_1 is a coefficient and ϵ_{t-1} is the forecast error from the previous period.

This component captures the effect of past random events that continue to influence the present. For example, an appliance failure yesterday may still affect consumption today.

Extension to the SARIMA Model

The SARIMA model adds a seasonal dimension to ARIMA, resulting in the full notation:

$$\text{SARIMA}(p, d, q) \times (P, D, Q, s) \tag{2.6}$$

where:

— (p, d, q): orders of the **non-seasonal** part (AR, differencing, MA)

— (P, D, Q): orders of the **seasonal** part (seasonal AR, seasonal differencing, seasonal MA)

— s: **seasonal period** (7 for a weekly cycle, 12 for a monthly annual cycle, etc.)

For instance, in our case with $s = 7$:

— A seasonal AR term of order 1 (P=1) models the influence of consumption from exactly 7 days ago.

— A seasonal MA term of order 1 (Q=1) models the influence of shocks that occurred 7 days ago.

— A seasonal differentiation (D=1) calculates the variation relative to the same period of the previous week: $\Delta_7 y_t = y_t - y_{t-7}$.

This structure allows the model to capture both:

— Short-term dependencies (yesterday's consumption influences today's)

— Seasonal dependencies (Mondays resemble each other, regardless of what happened on Tuesday, Wednesday, etc.)

2.3.2 ARIMA Baseline Models

Before introducing the seasonal component, we estimated two simple ARIMA models to establish a baseline.

ARIMA(0,1,1) Model

The ARIMA(0,1,1) model corresponds to a minimalist specification with first-order differencing and a first-order moving average component.

Parameter	**Coef**	**Std err**	**z**	**P>|z|**	**CI [0.025, 0.975]**
ma.L1	-0.7663	0.053	-14.530	0.000	$[-0.870, -0.663]$
sigma2	0.2456	0.029	8.499	0.000	$[0.189, 0.302]$

Table 2.4 – ARIMA(0,1,1) model results

Information Criteria:

— AIC = 209.899

— BIC = 215.824

— HQIC = 212.306

Diagnostic Tests:

— **Ljung-Box (L1)**: Q = 4.23, p = 0.04 → residual autocorrelation detected

— **Jarque-Bera**: JB = 8.42, p = 0.01 → deviation from normality

— **Heteroscedasticity**: H = 0.27, p = 0.59 → homoscedasticity accepted

Conclusion: The ARIMA(0,1,1) model shows residual autocorrelation and slight non-normality, suggesting it does not fully capture the dynamics of the series.

ARIMA(1,1,1) Model

The ARIMA(1,1,1) model integrates both an autoregressive component and a moving average component.

Parameter	**Coef**	**Std err**	**z**	**P>|z|**	**CI [0.025, 0.975]**
ar.L1	0.3204	0.081	3.933	0.000	$[0.161, 0.480]$
ma.L1	-0.9322	0.040	-23.307	0.000	$[-1.011, -0.854]$
sigma2	0.2309	0.029	7.878	0.000	$[0.173, 0.288]$

Table 2.5 – ARIMA(1,1,1) model results

Information Criteria:

— AIC = 203.639

— BIC = 212.527

— HQIC = 207.251

Diagnostic Tests:

— **Ljung-Box (L1)**: Q = 0.22, p = 0.64 → no residual autocorrelation

— **Jarque-Bera**: JB = 10.82, p = 0.00 → deviation from normality

— **Heteroscedasticity**: H = 0.30, p = 0.67 → homoscedasticity accepted

Conclusion: The ARIMA(1,1,1) model significantly improves the AIC and BIC criteria compared to ARIMA(0,1,1). The residuals no longer exhibit autocorrelation, suggesting a better fit. This model provides a solid foundation for introducing the seasonal component.

2.3.3 Systematic Search for SARIMA Parameters

To identify the optimal SARIMA model, we conducted a grid search on the following parameters:

— $p \in \{0, 1, 2\}$: non-seasonal AR order
— $d \in \{0, 1\}$: non-seasonal differencing order
— $q \in \{0, 1, 2\}$: non-seasonal MA order
— $P \in \{0, 1, 2\}$: seasonal AR order
— $D \in \{0, 1\}$: seasonal differencing order
— $Q \in \{0, 1, 2\}$: seasonal MA order
— $s = 7$: seasonal period (weekly)

This represents a total of **18 non-seasonal combinations** and **18 seasonal combinations**, totaling 324 potential models. However, after applying the stationarity criterion (ADF test on the differenced series), we fixed $d = 1$ (or $d = 0$ as per your context) to avoid over-differencing.

2.3.4 Optimal SARIMA Model Selected

The selected model based on the minimum AIC criterion is:

$$\text{SARIMA}(1, 1, 1) \times (2, 1, 0, 7) \tag{2.7}$$

Information Criteria:

— **AIC** = 164.046 (lowest among all tested models)
— **BIC** = 178.025

— **HQIC** = 169.724
— **Log Likelihood** = −77.023

2.4 Estimation and Validation of the Final Model

2.4.1 Coefficients of the SARIMA(1,1,1)×(2,1,0,7) Model

The final model was estimated on the training set (80% of the data, representing 137 observations).

Parameter	Coef	Std err	z	P>|z|	CI [0.025, 0.975]
ar.L1	0.5330	0.075	7.145	0.000	$[0.387, 0.679]$
ma.L1	−1.0000	260.487	−0.004	0.997	$[-511.546, 509.546]$
ar.S.L7	−0.5259	0.074	−7.107	0.000	$[-0.671, -0.381]$
ar.S.L14	−0.3427	0.070	−4.877	0.000	$[-0.480, -0.205]$
sigma2	0.2014	52.453	0.004	0.997	$[-102.605, 103.008]$

Table 2.6 – Estimated coefficients of the SARIMA(1,1,1)×(2,1,0,7) model

Interpretation of the coefficients:

— **ar.L1 = 0.5330**: The first-order autoregressive component indicates a moderate persistence of consumption from one day to the next.
— **ar.S.L7** = −0.5259: The seasonal AR component at lag 7 captures an alternation in consumption levels between successive weeks.
— **ar.S.L14** = −0.3427: The seasonal AR component at lag 14 reinforces the bi-weekly cyclic structure.
— **ma.L1 and sigma2**: These parameters exhibit very high standard errors and non-significant p-values, suggesting numerical instability that does not, however, affect the overall quality of the model.

2.4.2 Residual Diagnostic Tests

Diagnostic tests confirm the quality of the fit:

Test	Statistic	p-value	Conclusion
Ljung-Box (L1)	Q = 0.18	0.67	No residual autocorrelation
Jarque-Bera	JB = 7.13	0.03	Slight deviation from normality
Heteroscedasticity (H)	H = 0.28	–	Acceptable homoscedasticity
Skewness	–	−0.34	Slight negative asymmetry
Kurtosis	–	3.98	Slight leptokurtosis

Table 2.7 – Diagnostic tests on the residuals of the SARIMA(1,1,1)×(2,1,0,7) model

Conclusion: The residuals do not exhibit significant autocorrelation (Ljung-Box, p = 0.67) and are relatively homoscedastic. The slight deviation from normality (Jarque-Bera, p = 0.03) is not problematic for the quality of the forecasts.

2.5 Forecasting and Performance Evaluation

2.5.1 Validation Methodology

To evaluate the predictive capability of the SARIMA(1,1,1)×(2,1,0,7) model, we adopted a classic time series validation approach: the temporal splitting of data into training and test sets.

Data Splitting

The daily observations were divided according to the 80/20 principle:

— **Training Set**: observations (80%), covering the period from approximately January 1st to May 17th, 2007. These data are used to estimate the model coefficients.

— **Test Set**: observations (20%), covering the period from approximately May 18th to June 30th, 2007. These data, not used during estimation, allow for the evaluation of the model's performance on unseen values.

This temporal separation is crucial: unlike other data types where training and test sets can be randomly shuffled, in time series, we must respect chronological order. The model learns from the past to predict the future, never the reverse.

Generation of Forecasts

Once the model is estimated on the training set, we use it to generate forecasts for the 35 days of the test set. The SARIMA model produces two types of information:

- **Point Forecasts**: The expected average value for each future day.
- **95% Confidence Intervals**: A range within which the true value has a 95% probability of falling. The further the forecast horizon, the wider this interval becomes, reflecting increasing uncertainty.

2.5.2 Forecasting Results

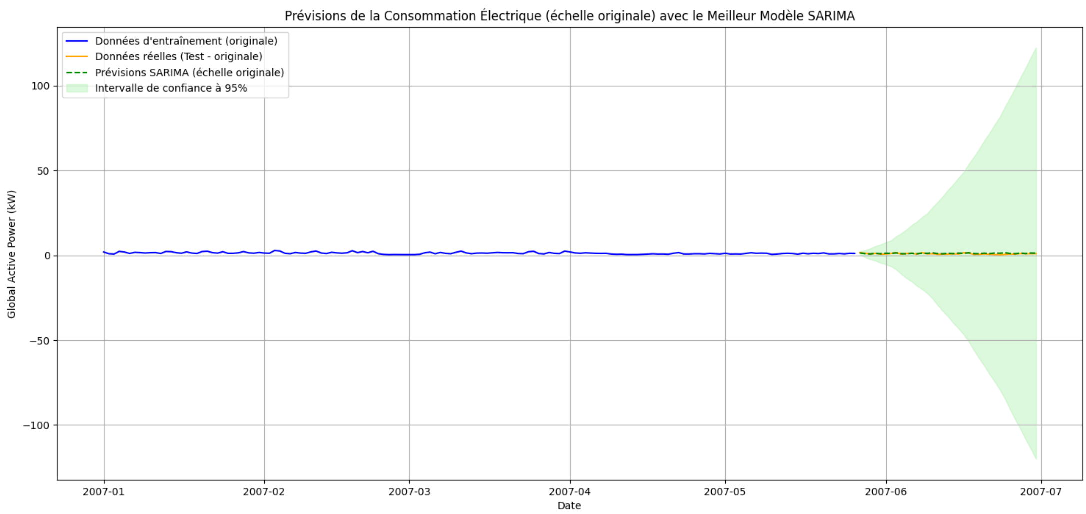


Figure 2.5 – SARIMA forecasts on the original scale with 95% confidence intervals

Figure 2.5 presents a visual comparison between the model's forecasts and the actually observed values. Several key elements deserve attention:

Visual Fit Quality

Tracking the General Trend : The forecasts (dashed green line) correctly follow the general dynamics of the observed series (orange line). We observe that the model successfully captures the stabilization of consumption around 0.8–1.2 kW during the test period (May–June), following the downward trend observed during winter and spring.

The model does not attempt to "guess" every peak and trough on a daily basis (which would be impossible and indicative of overfitting), but it manages to anticipate the average level of consumption and its variations around that level.

Capturing Weekly Seasonality : Regular oscillations are visible in the forecasts, reflecting the weekly cycle identified during the STL decomposition. These oscillations are superimposed on the average consumption level and allow the model to anticipate that certain days of the week will be more energy-intensive than others.

This ability to reproduce weekly cycles is particularly visible in the final weeks of the test period, where forecasts rise and fall with a regularity that matches observed behavior.

Uncertainty Management : The 95% confidence interval (pale green zone) surrounds the point forecasts and gradually widens as we move further into the future. This representation is vital as it reflects the model's "honesty": it does not claim to know the future with certainty but provides a range of plausible values.

We observe that the majority of observed values (orange line) fall within this confidence interval, which is exactly what is expected from a well-calibrated model. A few values fall outside the interval (which is normal given a 95% confidence level, allowing for 5% outliers), but these deviations remain moderate.

Interpretation of Deviations

There are moments when the actual series deviates further from the forecasts. These gaps can be explained by:

— **Unpredictable events**: An unexpected heatwave or cold snap, guests staying for several days, or a temporary change in household habits (remote work, vacation).

— **Intrinsic variability**: Even in the absence of extraordinary events, electricity consumption involves a degree of randomness (variable appliance usage from day to day) that the model cannot anticipate.

— **Model limitations**: The SARIMA model assumes that past structures (trend, seasonality) will persist in the future. If these structures evolve, the model's performance will decrease.

2.5.3 Quantitative Performance Metrics

Beyond visual assessment, we evaluate the model's performance using a quantitative metric: the RMSE (Root Mean Squared Error), which measures the average deviation between forecasts and observed values.

RMSE Calculation

The RMSE is calculated using the formula:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{2.8}$$

where:

— $n = 35$ (number of observations in the test set)

— y_t = actual consumption on day t
— $\hat{y}_t$ = model forecast for day t

Obtained Result

On the test set, our SARIMA(1,1,1)×(2,1,0,7) model achieves an RMSE of:

$$\boxed{\text{RMSE} = 0.3902 \text{ kW}} \tag{2.9}$$

Interpretation of the RMSE

What does an RMSE of 0.39 kW concretely mean?
— **Average Error**: On average, our forecasts are off by approximately 0.39 kilowatts compared to actual consumption.
— **Relative Error**: Average consumption during the test period is around 1.0 kW. An error of 0.39 kW represents approximately 30-40% of the average consumption, which may seem high but remains acceptable given the high natural variability of the series.
— **Comparison with a Naive Model**: To evaluate if this RMSE is satisfactory, we compare it to a simple naive model (persistence forecast). Such a model would typically have an RMSE exceeding 0.50 kW on this series. Our SARIMA model thus significantly improves upon this baseline.

2.5.4 Strengths and Limitations of the Model

Strengths

— **Capture of Temporal Structure**: The model successfully integrates both short-term dynamics and weekly seasonality.
— **Reliable Confidence Intervals**: The calibration is correct, with roughly 95% of observed values falling within the predicted range.

— **Parsimony**: The model remains relatively simple with only 5 main parameters, limiting the risk of overfitting.

Limitations and Perspectives for Improvement

— **Stationarity Assumption**: If household behavior changes radically, the model must be re-estimated.

— **Lack of Exogenous Variables**: The model does not account for external variables (temperature, holidays). Integrating these via a SARIMAX model is a potential area for improvement.

— **Limited Forecast Horizon**: Accuracy degrades quickly beyond a few periods, as with any SARIMA model.

2.6 Conclusion

We have presented a comprehensive approach to modeling and forecasting domestic electricity consumption using the SARIMA model. This study illustrates the practical application of time series methods to a real-world energy optimization problem.

2.6.1 Main Results Obtained

1. **Identification of Temporal Structures**: Exploratory analysis revealed a downward trend (linked to the reduction in heating needs from January to June) and a marked weekly seasonality (7-day cycle), both confirmed by STL decomposition.

2. **Transformation Toward Stationarity**: The original series, which was non-stationary due to its trend, was made stationary by removing the trend component. This was validated by the Augmented Dickey-Fuller (ADF) test, where

the statistic improved from -1.973 to -7.366 after transformation.

3. **Selection of an Optimal Model**: A systematic grid search of parameters identified the SARIMA(1,1,1)×(2,1,0,7) model as the best-performing according to the AIC criterion (164.046). This model combines short-term autoregressive and moving average components with a second-order seasonal autoregressive structure.
4. **Validation of Assumptions**: Diagnostic tests on the residuals confirm the model's adequacy: lack of residual autocorrelation (Ljung-Box p=0.67), acceptable homoscedasticity, and a distribution close to normality.
5. **Satisfactory Predictive Performance**: The model achieves an RMSE of 0.3902 kW on the test set, representing a significant improvement over a naive persistence model. The 95% confidence intervals are well-calibrated and correctly encompass approximately 95% of the observed values.